



Scientific Committee on Consumer Safety

SCCS

SCIENTIFIC ADVICE ON

hair dye 'Basic Brown 16'

(C009)

(CAS No. 26381-41-9, EC No. 247-640-9)

Submission V ter.



The SCCS adopted this document on 2 February 2026

ACKNOWLEDGMENTS

SCCS members listed below are acknowledged for their valuable contribution to the finalisation of this scientific advice.

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This Scientific Advice has been subject to a commenting period of min eight weeks after its initial publication (from 17 November 2025 to 19 January 2026, due to holiday period). Comments received during this period were considered by the SCCS. No change in the final text was made.

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SCCS

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1. ABSTRACT

The SCCS concludes the following:

1. *In light of the data provided, does the SCCS consider Basic Brown 16 safe, when used as an ingredient at 2 % in non-oxidative hair dye formulations?*

Having assessed the data provided, the SCCS does not consider the use of Basic Brown 16 safe when used as an ingredient in non-oxidative hair dye formulations because the weight of evidence indicates its potential for mutagenicity.

2. *Does the SCCS have any further scientific concerns with regard to the use of Basic Brown 16 in cosmetic products?*

/

Keywords: SCCS, scientific advice, hair dye, Basic Brown 16, C009, 8-[(4-Aminophenyl)diazonyl]-7-hydroxy-N,N,N-trimethylnaphthalen-2-aminium chloride, CAS No. 26381-41-9, EC No. 247-640-9, submission V ter., Regulation 1223/2009

Opinion to be cited as: SCCS (Scientific Committee on Consumer Safety), Scientific Advice on hair dye 'Basic Brown 16' (C009) (CAS No. 26381-41-9, EC No. 247-640-9) – submission V ter., preliminary version of 30 October 2025, final version of 2 February 2026, SCCS/1684/25

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2. MANDATE FROM THE EUROPEAN COMMISSION

Background

The ingredient with the INCI name hair dye 'Basic Brown 16' and chemical name '8-[(4-Aminophenyl)diazenyl]-7-hydroxy-N,N,N-trimethylnaphthalen-2-aminium chloride' (CAS No. 26381-41-9, EC No. 247-640-9) may be used as an ingredient in non-oxidative hair dye formulations.

The Scientific Committee on Consumer Products (SCCP) issued an opinion (SCCP/1165/08) in September 2008 with the conclusions that some aspect related to the genotoxicity potential of this Basic Brown 16 needed clarification. In April 2012, industry submitted another dossier (submission IV) describing the results of two *in vitro* genotoxicity studies together with a safety assessment for the use of this ingredient. The SCCS considered that (a) a conclusion on the mutagenic potential of Basic Brown 16 could not be drawn and (b) a characterisation of impurities in Basic Brown 16 and data on its stability in test solutions were required. The next submission (i.e., submissions V and V bis), where transmitted in 2013 and 2023, respectively, but could not be evaluated.

With the current submission (i.e., submission V ter.), received in end of September 2024, the applicant requests to assess the safety of Basic Brown 16 in view of the newly provided information on genotoxicity studies and complementary mutagenicity tests in mammalian cells, when used as an ingredient at 2 % in non-oxidative hair dye formulations.

Terms of reference

1. *In light of the data provided, does the SCCS consider Basic Brown 16 safe, when used as an ingredient at 2 % in non-oxidative hair dye formulations?*
2. *Does the SCCS have any further scientific concerns with regard to the use of Basic Brown 16 in cosmetic products?*

3. SCIENTIFIC ADVICE

3.1 Physicochemical specifications

3.1.1 Chemical identity

3.1.1.1. Primary name and/or INCI name

Basic Brown 16 (INCI)

3.1.1.2 Chemical names

IUPAC: 8-[(4-aminophenyl)diazenyl]-7-hydroxy-N,N,N-trimethylnaphthalen-2-aminium chloride

Ref.: ECHACHEM database <https://chem.echa.europa.eu/100.043.295/dossier-list/reach/dossiers/active?searchText=26381-41-9>

From SCCS/1496/12 Opinion

2-Naphthalenaminium, 8-[(4-aminophenyl)azo]-7-hydroxy-N,N,N-trimethyl-, chloride

Ammonium,[8-[(p-aminophenyl)azo]-7-hydroxy-2-naphthyl]-trimethyl-, chloride

8-[(4-Aminophenyl)diazenyl]-7-hydroxy-N,N,N-trimethylnaphthalen-2-aminium chloride

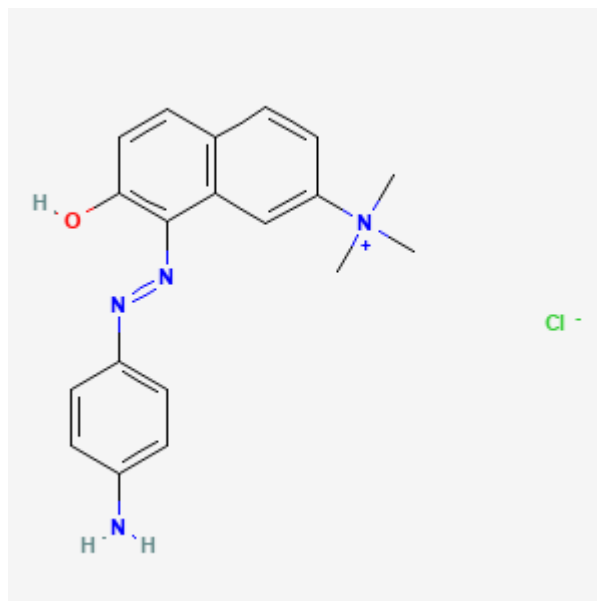
Ref.: SCCS/1496/12 Revision 26 February 2013

3.1.1.3 Trade names and abbreviations

Arianor Mahogany
Arianor Mahogany 306002
C.I. 12250
C.I. Basic Brown 16
COLIPA C 009

3.1.1.4 CAS / EC number

CAS No. 26381-41-9
EC No. 247-640-9

3.1.1.5 Structural formula

Ref. : <https://pubchem.ncbi.nlm.nih.gov/compound/Basic-brown-16> (consulted on the 8th April 2025)

3.1.1.6 Empirical formula

Molecular Formula
 $C_{19}H_{21}N_4O.Cl$

Ref.: <https://pubchem.ncbi.nlm.nih.gov/compound/Basic-brown-16> (consulted on the 8th April 2025)

3.1.2 Physical form

Dark green to black powder

3.1.3 Molecular weight

356.8 g/mol

Computed by PubChem 2.2 (PubChem release 2024.11.20)

Ref.: <https://pubchem.ncbi.nlm.nih.gov/compound/Basic-brown-16> (consulted on the 8th April 2025)

3.1.4 Purity, composition and substance codes

The purity and composition of different batches is reported in the following section.

3.1.1 Impurities / accompanying contaminants

According to the Applicant, the test substance is one main component (see structure 3.1.1.5). Few impurities could be present and their characterization has been done through a HPLC

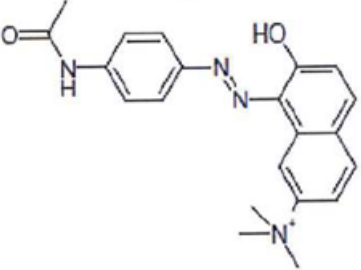
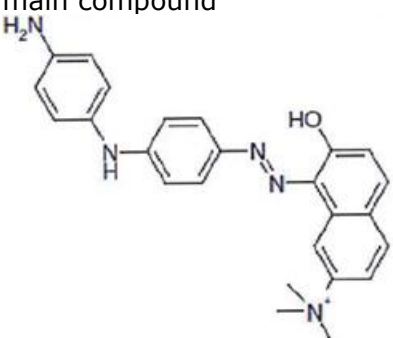
with UV/Vis detection and ESI-MS analysis. A structure could be assigned to the peaks that are quantified by area% detected at 260 and 350-550 nm.

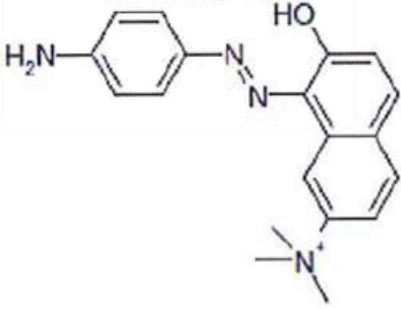
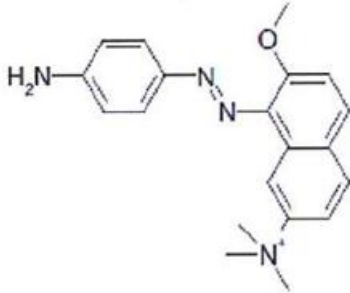
The purity of the main component was 99.54 area-% at 260 nm and 99.56 area-% at 350-550 nm. The remainder of the sample consists of minor impurities, each of them being present at <1.0% as well as compounds which are not considered to contribute to the desired coloration effect. They represent 0.47 area% at 260 nm and 0.44-area% at 350-550 nm.

Chloromethane and bromomethane has been determined by Headspace GC-MS, p-phenylenediamine has been determined by LC-MS/MS.

In the following Tables the observed peaks and their assigned structure are given together with the corresponding area-% for the batch 95424101.

Table 1. Summary of LC/DAD/MS analysis of Basic Brown 16-batch 95424101

RT (min)	m/z (positive mode)	Remarks, proposed structure	Area% at 260 nm	Area % at 350-550 nm
9.66	363	Molecular weight (MW) 42 mass units higher than MW of the main compound  Molecular Weight = 363.4 Monoisotopic mass = 363.2 Molecular formula= C21H23N4O2	0.15	0.14
11.17	412	MW 91 mass units higher than MW of the main compound  Molecular Weight = 412.5 Monoisotopic mass = 412.2 Molecular formula= C25H26N5O	0.12	0.08

11.34	321	Main compound  Molecular Weight = 321.4 Monoisotopic mass = 321.2 Molecular formula= C ₁₉ H ₂₁ N ₄ O	99.54	99.56
17.56	412	Isomer of compound at 11.17 min	0.09	0.08
22.68	335	MW 14 mass units higher than MW of the main compound e.g.  Molecular Weight = 335.4 Monoisotopic mass = 335.2 Molecular formula= C ₂₀ H ₂₃ N ₄ O	0.11	0.14

Other potential impurities (Batch 95424101)

ID	95424101 (ppm)
p-phenylenediamine	9.5
chloromethane	Below limit of quantitation 0.20 estimated
bromomethane	Below limit of detection <0.1

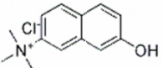
Quantification of p-phenylenediamine (PPD) impurity in various batches

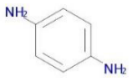
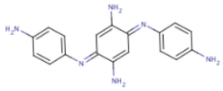
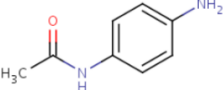
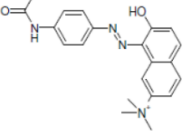
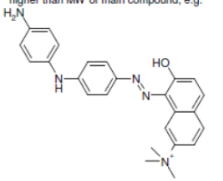
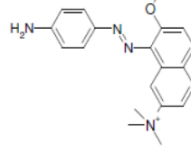
The samples of different batches of the test substance were dissolved in water containing 0.1% HCOOH. Afterwards the solutions were immediately analyzed by UPLC-MS/MS with defined concentrations of freshly prepared PPD standards (multipoint calibration).

Sample Basic Brown 16	Content of PPD in ppm
Batch 40	15.1
Batch 43	11.6
Batch 95424101	9.5
Batch 57861/2	15.3

The purity and composition of different batches is reported below.

Table 3. Composition of 5 batches

		Batch				
	Structure	57861/2	KS6054	95424101	12/13	VW230921
Identity: verified by		NMR, IR and UV	IR and UV	NMR, IR and UV	NMR, IR and UV	NMR, IR and UV
Purity by NMR assay		89.1 (W/W)	65.7%	88.6%	65.7%	90.0%
Purity by HPLC		99.5%		99.5%	98.9 - 97.9	99.6%
Methylchloride 74-87-3		<0.15 ppm 2%	nr	0.20 ppm (est.)	2.9%	nr
Methylbromide 74-83-9		<0.1 ppm 1%	nr	< 0.1 ppm	nr	nr
NBTRI	 MW: 237.5 as chloride MW: 313 as methyl sulfate	ND 0.13% (or 0.13/89.1=0.15%)	0.9% (or 0.9/60=1.5%)	ND	nr	ND

PPD 106-50-3		15.3 ppm	na	9.5 ppm 17 ppm	nr	nr
Bandrowski's Base 20048-27-5		nr	nr	5 ppm	nr	nr
4- aminoaceta nide 122-80-5	Molecular Weight:150 	nr	nr	2 ppm	nr	ND
Compound B	Molecular weight (MW) 42 mass units higher than MW of main compound e.g.  Molecular Weight =363.4 monoisotopic mass =363.2 Molecular Formula =C21H23N4O2	0.3% (260 nm) 0.26% (350-550 nm)	nr	0.15% (260 nm) 0.14% (350-550 nm)	nr	0.42
Compound C	Molecular weight (MW) 91 mass units higher than MW of main compound, e.g.  Molecular Weight =412.5 monoisotopic mass =412.2 Molecular Formula =C25H26N5O	0.07% (260 nm) 0.05% (350-550 nm)	nr	0.21% (260 nm) 0.16% (350-550 nm)	nr	0.10
Compound E	MW 14 mass units higher than MW of main compound e.g.  Molecular Weight =335.4 monoisotopic mass =335.2 Molecular Formula =C20H23N4O	0.08% (260 nm) 0.09 (350-550 nm)	nr	0.11% (260 nm) 0.14%(350-550 nm)	nr	ND
Chloride: by ion chromatogra phy		11.5 (W/W)		12.5% (W/W)	20.3 (W/W)	11.6 (W/W)

Bold: recent analysis for this Submission Vbis / normal: previous analysis available in SCCS Opinion (SCCS/1496/12) values

nr: not reported values

ND: Not Detectable

PPD: p-phenylenediamine

Ref.: SENSIENT & VIVIMED. Glossary of compounds present within Hair Dye C009.
Report 12250_5. 2013.

Table 4. Chemical composition of global market material

<u>Purity</u>	
Assay by NMR	> 66 weight-%
Chloride	< 21.0 weight-%
Sucrose	< 25.0 weight-%
Sulphated ash	≤ 30.0 weight-%
Solvent content (water)	< 6.0 weight-%
<u>Purity by HPLC assay:</u>	
Basic Brown 16	> 96 area-%
Compound B	< 1.0 area-% (MW 363)
Compound C	< 0.3 area-% (MW 412)
Compound E	< 1.5 area-% (MW 335)
NBTRI (7-Hydroxy-N,N,N-trimethylnaphthalen-2-aminium chloride)	< 0.5 area-% (MW 237.5)
4-aminoacetanilide	< 10 ppm (MW 150)
Methylbromide*	< 0.1 ppm
Methylchloride*	< 0.4 ppm
p-phenylenediamine	< 20 ppm
Heavy Metals (ppm)	Pb<20, Sb<10, Ni<10, As<5, Cd<5, Hg<1

* Methylbromide: Harmonized classification under CLP. Index number: 602-002-00-2 (Muta 2)

* Methyl Chloride: Harmonized classification under CLP. Index number: 602-001-00-7 (Carc 2 and CoRap list)

SCCS comments on purity and impurity

In the current submission, quantitation of purity and chemical characterisation of the impurities was performed in batches 57861/2 and 95424101 using HPLC on a C18 (150 x 4.6 mm, 3.5 µm) analytical column with PDA detection (at 260nm and 350-550nm) and ESI/MS analysis. In these batches, chloromethane and bromomethane were determined by Headspace GC-MS and p-phenethylamine by UPLC/MS/MS. However, detailed data on the chemical characterisation of the impurities by LC-MS was submitted only for batch 95424101. For batch VW230921A, the analysis was conducted using a different HPLC system (Phenyl-hexyl column, 2.1 × 100 mm), and detection was performed at 260 nm and 476 nm. No LC-MS data were provided to confirm identity of the impurities reported in Table 3.

While the main constituent and impurities of Basic Brown 16 have been more precisely characterised in batch 95424101, the overall impurity content is higher in the proposed specification for global market material (Table 4) compared to the batch used in the genotoxicity studies (VW230921A, Table 3). In the VW230921A batch, impurity B was present at 0.42%, impurity C at 0.10%, and impurity E was not detected, whereas the proposed specification for global market material allows up to 1% for B, 0.3% for C, and 1.5% for E.

The increased impurity levels (up to 2.8%-area, Table 4) raise uncertainty in regard to applicability of the provided toxicological data. Therefore, without further justification, the toxicological studies on VW230921A cannot be considered representative of the material covered under the proposed specifications.

Additionally, the impurity concentration values and profile of batch VW230921A reported in Table 3 are not consistent with those reported in the corresponding analytical file submitted along with the dossier.

3.1.5 Solubility

From the Applicant

Water: 373 g/L at 20°C EU Method A.6
 Ethanol: 10-100 g/L at room temperature
 DMSO: 50-200 g/L at room temperature

SCCS comment on solubility

Basic Brown 16 is freely soluble in water.

3.1.6 Partition coefficient (Log P_{ow})

From the Applicant

Log Pow: -0.74 (pH 9.95, 23°C) EU Method A.8

3.1.7 Additional physicochemical specifications

From the Applicant

Appearance: Dark green to black powder
 Melting point: 169-175 °C (in SCCNFP/0668/03 it was 160-170 °C-decomposition)
 Flash point: /
 Vapour pressure: /
 Density: /
 Viscosity: /
 pKa: /
 Refractive index: /
 pH: /
 UV_Vis spectrum: (200-800 nm): λ_{max} 218 nm, 259 nm, 478 nm

Ref.: SENSIENT COSMETIC TECHNOLOGIES. Characterization of impurities in batch VW230921A used in toxicological tests. 2023.

From ECHA <https://echa.europa.eu/el/registration-dossier/-/registered-dossier/17386/4/2> (consulted on 27/04/2025):

Odour: Odourless
 Boiling point: 644.52°C at 101 325 Pa (Danish QSAR modelling database, 2016)
 Density: 1.2 g/cm³
 Particle size distribution: 150 - 10 µm (granulometric analysis with rotap sieve shaker)
 Flash point: 398.1 °C at 101 325 Pa

3.1.8 Homogeneity and Stability

From the Applicant

A freshly prepared sample of Basic Brown 16 – batch 95424101 was compared by LC-DAD with a sample stored for 72 h at autosampler conditions (4°C). No decrease of the main component was observed. According to the integrals of the main peak the sample is stable within a time frame of 72 h at 4°C, as reported in the following table. This batch is the one used in the Comet assay.

Ref.: SENSIENT COSMETIC TECHNOLOGIES. Characterization of impurities in batch 95424101 used in toxicological tests and stability. 2023.

Table 5. Comparison of main peak of a freshly prepared and stored solution

Batch 95424101	
Time Point 0h	99.49
Time Point 72h	99.46
Recovery (Time Point 72h / Time Point 0h)	100.0%

General comments by the Applicant on physicochemical characterisation

The different comments raised in the previous opinion of the SCCS about impurities and stability have been addressed above.

The analytical analysis is available for the 3 batches used in the toxicological tests (57861/2, 95424101 and VW230921).

Refs.: SENSIENT COSMETIC TECHNOLOGIES. Characterization of impurities in batch 95424101 used in toxicological tests and stability. 2023.

SENSIENT COSMETIC TECHNOLOGIES. Characterization of impurities in batch VW230921A used in toxicological tests. 2023.

SENSIENT & VIVIMED. Characterization of impurities in batch 57861/2 used in toxicological tests. 2013.

3.2 Function and uses

From the Applicant

Basic Brown 16 is used as a direct dye for hair colouring products without mixing with an oxidising agent (e.g. hydrogen peroxide). The final concentration on head of Basic Brown 16 can be up to 2.0%, in about 100 ml, about once per month.

3.3 Toxicological evaluation

3.3.1. Acute toxicity

3.3.1.1. Acute oral toxicity

From SCCS/1496/12:

"The acute median lethal oral dose (LD50) of Basic Brown 16 was reported to be between 2 and 4 g/kg bw."

3.3.1.2. Acute dermal toxicity

/

3.3.1.3. Acute inhalation toxicity

/

3.3.1.4. Acute intraperitoneal toxicity

/

3.3.2. Irritation and corrosivity**3.3.2.1. Skin irritation**

From SCCS/1496/12:

"Basic Brown 16 was considered not to be irritating to rabbit skin."

3.3.2.2. Mucous membrane irritation / Eye irritation

From SCCS/1496/12:

"Basic Brown 16 showed irritant potential of the rabbit eye."

3.3.3. Skin sensitisation

From SCCS/1496/12:

"Basic Brown 16 was a moderate skin sensitizer under the conditions of the test. An EC3 value of 12.2% was derived."

3.3.4. Dermal / percutaneous absorption

From SCCS/1496/12:

"The amount considered to be absorbed as direct dye formulation was 5.15 µg/cm² (range 2.19 to 9.12) or 1.42% of the applied dose (range 0.58% to 2.91%). At 2% dissolved in water, the amount considered to be absorbed was 4.57 µg/cm² (range 3.76 to 5.80) or 1.08% of the applied dose (range 0.89% to 1.38). As too few test chambers (8 chambers from 2 donors) the mean + 2 SD (5.15 + 2 × 2.32 = 9.79 µg/cm²) may be used for calculating the MOS."

3.3.5. Repeated dose toxicity**3.3.5.1. Repeated Dose (14 days) oral toxicity**

/

3.3.5.2. Sub-chronic (90 days) toxicity (oral)

From SCCS/1496/12:

"In a 90-day rat study, the NOAEL was considered to be 50 mg/kg bw/day, based on decreased body weight gain and some neurotoxic symptoms in both sexes. This value was re-calculated to 33 mg/kg bw/day based on the dye content of the test substance. The NOAEL both for maternal and embryo-foetal toxicity was considered to be 90 mg/kg bw/day. This value was re-calculated to 59 mg/kg bw/day based on the dye content of the test substance."

3.3.5.3. Chronic (> 12 months) toxicity

/

3.3.6. Mutagenicity / Genotoxicity

3.3.6.1. Mutagenicity / Genotoxicity *in vitro*

From the Applicant

From SCCS/1496/12:

"Overall, the genotoxicity of Basic Brown 16 was investigated for the three endpoints of genotoxicity: gene mutations, chromosome aberrations and aneuploidy.

Basic Brown 16 did induce gene mutations in bacteria. In mouse lymphoma cells, the mutant frequency at both the tk as well as at the hpert locus was increased. In a hpert gene mutation test in V79 cells with limited value, an increase in the mutant frequency was not observed. The increase in the mutant frequency at the tk locus of the mouse lymphoma cells was due to an increase in small colonies, which in turn is indicative for a clastogenic effect of Basic Brown 16. The positive *in vitro* micronucleus test confirmed the clastogenic potency of Basic Brown 16. A Comet assay, indicative for both clastogenicity and mutagenicity, performed with reconstructed human skin was negative. Clastogenicity found *in vitro* could not be confirmed *in vivo*. Basic Brown 16 did not induce micronuclei in an *in vivo* micronucleus test in erythrocytes of mice. The positive finding in the gene mutation tests in bacteria and mammalian cells were not confirmed nor overruled with an *in vivo* test measuring the same genotoxic endpoint. Instead of an *in vivo* gene mutation test a Comet assay, indicative for both clastogenicity and mutagenicity, in reconstructed human skin was performed. This *in vitro* model strongly resembles human skin and may be considered as a justified surrogate for an *in vivo* gene mutation test. The absence of DNA damage in this Comet in reconstructed human skin may indicate that the chance that Basic Brown 16 has a genotoxic potential is low. At present, this test model is not sufficiently validated. A final decision on the mutagenic potential cannot be made at the moment".

The SCCS note

With the new submission V ter. the Applicant provided results of one study on gene mutations in L5178Y cells (*Hprt* locus) and one *in vitro* micronucleus test on human peripheral blood lymphocytes using VW230921A batch. Additionally, the Expert Opinion on the Genotoxicity Data was provided by the Applicant.

The results of both studies and the summary of the Expert's Opinion with the SCCS comments are summarised below.

New *in vitro* study #1

Gene Mutation test at the *hprt* Locus in L5178Y cells

Guideline: OECD Guideline 476 (2016)

Cells:	mouse lymphoma L5178Y cells
Replicates:	2 per treatment
Test substance:	BASIC BROWN 16
Solvent:	purified water
Batch:	VW230921A
Purity:	90.0% (1H-NMR)
Concentrations:	10, 25, 50, 75, 100, 125, 150, 200, 225, 250, 275, 300, 400, 500 µg/mL for 3h treatment without S9; and 10, 25, 50, 125, 150, 225, 250, 275, 300, 400, 500 µg/mL for 3h treatment with S9.
Treatment:	3-hour treatment with or without S9-mix
Expression period:	7 days
Positive controls:	-S9-mix: 4-nitroquinoline 1-oxide (NQO): 0.15 and 0.20 µg/mL +S9-mix: benzo(a)pyrene (B[a]P): 4 and 6 µg/mL
GLP:	yes
Study period:	February 13, 2024 - July 11, 2024

Methods

Basic Brown 16 was assayed for the ability to induce mutation at the hypoxanthine-guanine phosphoribosyl transferase (*hprt*) locus (6-thioguanine [6TG] resistance) in mouse lymphoma cells using a fluctuation protocol. The study consisted of a cytotoxicity Range-Finder Experiment followed by a Mutation Experiment, each conducted in the absence and presence of metabolic activation by a β -Naphthoflavone/Phenobarbital-induced rat liver post-mitochondrial fraction (S-9). The test article was formulated in purified water. A 3-hour treatment incubation period was used for each experiment.

Results

In the cytotoxicity Range-Finder Experiment, six concentrations were tested in the absence and presence of S-9, ranging from 62.5 to 2000 µg/mL (an acceptable maximum concentration for *in vitro* genetic toxicology studies according to current regulatory test guidelines). The highest concentration to give >10% relative survival (RS) was 250 µg/mL, which gave 14% and 19% RS in the absence and presence of S-9, respectively.

In the Mutation Experiment, fourteen concentrations, ranging from 10 to 500 µg/mL, were tested in the absence and presence of S-9. Seven days after treatment, the highest concentrations analysed to determine viability and 6TG resistance were 275 µg/mL in the absence of S-9 and 400 µg/mL in the presence of S-9, which gave 19% and 20% RS, respectively.

Vehicle and positive control treatments were included in the Mutation Experiment in the absence and presence of S-9. Mutant frequencies (MF) in vehicle control cultures fell within acceptable ranges and clear increases in mutation were induced by the positive control chemicals 4-nitroquinoline 1-oxide (NQO) (without S-9) and benzo(a)pyrene (B[a]P) (with S-9). Therefore, the study was accepted as valid.

When tested up to toxic concentrations for 3 hours in the absence and presence of S-9, statistically significant increases in mean MF, compared to the mean vehicle control values, were observed at 50, 100 and 150 to 275 µg/mL in the absence of S-9 and at 150 and 250 to 275 µg/mL in the presence of S-9 and a statistically significant linear trend was noted in the absence and presence of S-9. All statistically significant increases in mean MFs were above the historical vehicle control ranges generated by the last 20 experiments (mean $\pm$ 2 standard deviations) performed in this laboratory (0.23 to 8.26 mutants/10⁶ viable cells: mean value 4.24 in the absence of S-9 and 0 to 10.48 mutants/10⁶ viable cells: mean value 4.91 in the presence of S-9).

Results of formulation analyses demonstrated homogeneity and stability of the samples (5 hours at room temperature, protected from light) and achieved concentrations within 100 $\pm$ 10% of the nominal test article concentrations with an RSD of $\leq$ 5%, and were therefore considered acceptable. No significant degradation (>10%) was observed over the storage period. No test article was detected in the vehicle control sample.

Conclusion

It is concluded that Basic Brown 16 was mutagenic at the *hprt* locus in mouse lymphoma L5178Y cells when tested up to toxic concentrations for 3 hours in the absence and presence of a rat liver metabolic activation system (S-9), under the experimental conditions described.

Ref.: Nisar R. (Labcorp) for Sensient Beauty. Basic Brown 16: *In vitro* L5178Y Gene Mutation Assay at the *hprt* locus. 2024.

SCCS comment on the gene mutation study

The SCCS considers the study as valid and the results as clearly positive, both with and without metabolic activation at adequate concentrations (minimum Relative Survival of 15% without S9 or 20% with S9), fulfilling all 3 criteria for positivity recommended by OECD TG 476:

- at least one of the test concentrations exhibited a statistically significant increase compared with the concurrent negative control,
- the increase was concentration-related when evaluated with an appropriate trend test,
- some of the results were outside the distribution of historical negative control data.

New *in vitro* study #2

Micronucleus test on human peripheral blood lymphocytes

Guideline:	OECD Guideline 487 (2016), ICH S2(R1) (2011)
Cells:	human peripheral blood lymphocytes
Replicates:	4 wells per treatment (24 wells plates); final culture volume 500 µL
Test substance:	BASIC BROWN 16
Solvent:	sterile water for irrigation
Batch:	VW230921A
Purity:	99.6% HPLC
Concentrations:	5, 12.5, 25, 50, 125, 250, 500, 1250, 2500, 5000 µg/mL for 3h treatment 5, 12.5, 25, 50, 125, 250, 500, 1250, 2500, 5000 µg/mL for 24h treatment
Treatment:	3-hour treatment with S9-mix and 24-hour treatment without S9-mix.; addition of test article at 44 h after PHA stimulation; addition of Cyto B at initiation of exposure (at 44h for 24 h exposure or at 47 h after 3 h exposure; Cyto B for 24 h or 21 h); cell harvest time at 68 h after PHA stimulation
Positive controls:	+S9: Cyclophosphamide: 6.7 and 10 µg/mL -S9: Vinblastine: 0.01 µg/mL; mitomycin C: 0.2 µg/mL
GLP:	no (<i>comment by the Applicant</i> : The study was done in non GLP conditions. Results remain valuable. Between this non-GLP study and the recommendations of the OECD 487 guideline, limited criteria are different (mainly the 3-hour time without metabolic activation, not tested)
Study period:	February 28, 2022 - December 13, 2022

Methods

BASIC BROWN 16, CAS Number 26381-41-9, Batch number VW230921A, formulated in sterile water for irrigation, giving a dark brown solution, was tested in a range of concentrations from 5 to 5000 µg/mL in 3-hour treatment of human peripheral blood lymphocytes with S9-mix and 24-hour treatment without S9-mix.

Results

Cytotoxicity based on RI % value was marked and dose-related in both treatments, 24h long treatment and pulse 3h treatment in the presence of metabolic activation. No precipitation of BASIC BROWN 16 was observed whatever the tested concentration in both treatment conditions following cell culture incubation, however a dark coloration of supernatant was observed after treatment. Consequently, the concentrations selected to perform the micronucleus evaluations were 25, 50, 125 and 250 µg/mL for 3h treatment and 12.5, 25 and 50 µg/mL for 24h treatment, in accordance with the OECD recommendation i.e highest concentration of 5000 µg/mL in the absence of toxicity or with toxicity between 45 and 55 %.

Following BASIC BROWN 16 treatment of human peripheral blood lymphocytes, no biological significant increase in micronucleated binucleated cells frequency was observed neither after the 3-hour treatment with S9-mix nor after the 24-hour treatment in the absence of S9-mix. A statistically significant increase both in Fisher's exact and Bonferroni tests was observed at 25 µg/mL ($p \leq 0.01$) after 3-hour treatment with metabolic activation. However, this increase was not concentration-related, the value of micronucleated binucleated cells (9.9‰) was just higher than the maximum of the 95-percentile range of the historical negative control data (8.31‰) and lower than the minimum of the 95-percentile historical positive control ranges (11.10‰) and very close to the highest observed value of the historical negative control data (9.9‰ versus 9.7‰). Statistically significant increases only in Fisher's exact test ($p \leq 0.05$) were also observed in the 3h treatment with metabolic activation at 250 µg/mL and in the 24-hour treatment without metabolic activation at 12.5 µg/mL, these increases were not taken into account as Bonferroni's test was not significant for these concentrations and they were inside our historical negative data (95-percentile).

The most important, the Cochran-Armitage Linear Trend was not significant for the 3-hour and the 24-hour treatments with p-value of 0.104 and 0.286 respectively.

Negative and Positive controls met the acceptance criteria.

No biologically significant increase in micronucleated binucleated cells frequency was observed in both-treatment (3-hour and 24-hour). Statistically significant increase observed in 3-hour treatment at 25 µg/mL was judged as not relevant as just lower the upper value of the historical data and for an isolated concentration.

The criteria for a clear positive result are pair-wise statistical significance, trend test positive, and outside the historical data, the result of the 3-hour+S9 exposure meets 2 of these 3 criteria. In the case of an equivocal result, the OECD states:

"In case the response is neither clearly negative or clearly positive as described above and/or in order to assist in establishing the biological relevance of a result, the data should be evaluated by expert judgement and/or further investigations. Scoring additional cells (where appropriate) or performing a repeat experiment possibly using modified experimental conditions (e.g. concentration spacing, other metabolic activation conditions [i.e. S9 concentration or S9 origin]) could be useful."

However, based on the knowledge of this screening model, the very weak increases observed in table text did not show that BASIC BROWN 16 induced an increase in micronucleus frequency.

Conclusion

Under the experimental conditions of this screening study, BASIC BROWN 16 CAS Number 26381-41-9, Batch number VW230921A did not induce increase in micronucleus frequency in this present non GLP screening study.

Ref.: Cariou O. Basic Brown 16: *In vitro* Human Lymphocyte Micronucleus Assay, GenEvolutionN, France. Study number 3386-299. 2022.

SCCS comment on the micronucleus study

The SCCS considers the results of this non-standard and non-GLP study as equivocal for the following reasons:

- 3 h, -S9, condition of exposure was not used,

- both after 3 h, +S9, and 24 h, -S9, exposure showed slight, but significantly increased MN frequency at the lowest and the highest concentrations tested. The MN frequency of 9.9 ($p \leq 0.01$) at 25 $\mu\text{g/mL}$, +S9, was outside the control historical negative values.
- for unclear reasons, different total cell numbers were scored for MN in different exposure conditions. The lowest total cell number (~ 2000) was used when the highest MN frequency was observed (25 $\mu\text{g/mL}$, for 3 h, +S9 mix). Therefore, it cannot be excluded that a higher MN frequency would be observed if more cells had been scored.

3.3.6.2. Mutagenicity / Genotoxicity *in vivo*

No data was submitted.

Applicant's Opinion on the Genotoxicity Data

An independent review of the genotoxicity data for Basic Brown 16 (CAS# 26381-41-9) has been provided by the Applicant and is summarized below. The full review was provided to the SCCS as part of the dossier.

Table 6. Summary of the review by the Applicant of the genotoxicity data for Basic Brown 16:

- Available genotoxicity data:

Study Type	Result In vitro -S9	Result In vitro +S9	Result In vivo	Comment
Ames Test	Negative	Positive	NA	Reproducible but generally weak responses in TA98 and TA1537 with pre-incubation.
Micronucleus V79 cells	Equivocal	Positive	NA	Study may be unreliable because cytotoxicity was not evaluated correctly.
Micronucleus Lymphocytes	Negative	Negative	NA	A modern study with human relevant cells that have some metabolic competence provided clear negative results.
V79 hprt	Negative	Negative	NA	Study considered to be limited by SCCS because cytotoxicity was not evaluated correctly. However, the maximum dose +S9 exceeded the prescribed maximum, and the maximum dose -S9 was comparable with the hprt assay in L5178Y cells. The study is therefore reliable.
L5178Y hprt	Positive	Positive	NA	The positive result in the absence of S9 is unexpected because BB16 requires P450 activation and L5178Y cells are deficient in P450. This provides some concern on the validity of the study. The dose response relationship was not clear in both exposure conditions and both experiments.
L5178Y hprt	Positive	Positive	NA	The positive result in the absence of S9 is unexpected because BB16 requires P450 activation and L5178Y cells are deficient in P450. The dose response relationship was not clear in both exposure conditions and both experiments.
L5178Y tk	Equivocal	Negative	NA	Results with S9 contradict those for the same cell line at the hprt locus. The result without S9 is unexpected considering the metabolic deficiency of the cells. In the absence of S9 the cytotoxicity was at the upper limit of acceptability.
Epiderm Comet assay	Negative	NA	NA	A recent validation paper for the methodology supports the validity of this tissue relevant assay.
Micronucleus in mice	NA	NA	Negative	This is a robust study and shows no evidence of clastogenic or aneugenic potential <i>in vivo</i> .

- Conclusion of the full review:

The available study data provide evidence that BB16 may be metabolised to mutagenic metabolites in the bacterial mutation assay known as the Ames test. The Ames test is a highly sensitive assay and utilises a metabolic activation system (S9) that has pronounced P450 enzyme activity via the use of an enzyme induction regimen. The assay is useful for determining the potential of a substance to be mutagenic but cannot be used to confirm whether such activity is relevant to human exposure. There are several *in vitro* mammalian cell assay studies providing a complex mix of positive, equivocal and negative results that are difficult to rationalise into a coherent picture. However, it is relevant that in most of the studies there were technical deficiencies that reduce the reliability of the study conclusions; for example, incorrect measures of cytotoxicity that may have led to excessively toxic dose levels being used and spurious or false positive results (Scott *et al.*, 1991). Perhaps the most relevant of the *in vitro* studies is the most recent (not reviewed by the SCCS) micronucleus test in human lymphocytes performed in 2022 and this provided a clear negative result.

The *in vitro* bacterial and mammalian cell data provide some concern for both the potential mutagenicity and clastogenicity of BB16. However, many of the positive results reported in the mammalian cell assays are suspected false positives caused by excessive toxicity. The *in vivo* clastogenicity (and aneugenicity) potential of BB16 has been addressed by an historic

but robust *in vivo* micronucleus study in the mouse that used the IP dose route that maximised bioavailability and exposure to the target tissue; no evidence for such effects were observed. The mutagenic potential of BB16 could not be evaluated *in vivo* because of the cosmetic regulations of Europe. However, an *in vitro* human skin tissue model was used to evaluate the potential of BB16 to induce DNA damage using the Comet assay. The use of such models has recently been validated by a joint regulator, academia and industry group and the model has been shown to be highly predictive for *in vivo* mutagenicity.

Whilst some questions remain about the potential for BB16 to be genotoxic in humans the available body of evidence provides some assurance that, overall, the potential mutagenicity/genotoxicity of BB16 may be concluded to be not relevant to human exposure.

The SCCS evaluated a similar substance (BB17) in 2014 and concluded it to be safe for use in non-oxidative hair dye formulations. It seems that the very similar genotoxicity profiles for BB17 and BB16, (with the exception that the BB16 profile includes a negative *in vivo* study that is absent from the BB17 profile) indicate that both substances should receive safe for use conclusions by the SCCS.

Ref.: Expert Opinion on the Genotoxicity Data for Basic Brown 16. CEHTRA. 2024.

SCCS comments on the Applicant's Opinion on the genotoxicity data

The SCCS assessment of all available studies in the current and previous submissions with indication of differences in the provided assessments by the Applicant and by the SCCS is presented in Annex 1. To this Annex, the SCCS added 4 studies that were not considered by the Applicant: 1 Ames test, 1 chromosomal aberration test *in vitro*, 1 unscheduled DNA synthesis test *in vitro*, 1 micronucleus test *in vivo*. The SCCS could only find one L5178Y *Hprt* study which was mentioned in the Applicant's analysis. Although the Applicant mentioned a second such study, it appears to be a duplicate of the first.

The SCCS also considered that the proposed read-across between BB16 and BB17 is not justified because it does not fulfil the critical requirements for an unbiased and transparent read-across. For example, no details have been provided in regard to how the search for source analogues was carried out, how many were identified, and on what basis they were deselected or selected for use in the read-across. Therefore, although BB17 and BB16 share a structural similarity, the SCCS is of the opinion that read-across from a single selected analogue can only be considered relevant for use in risk assessments if it is not the only evidence for the given endpoint – i.e. there is further information from other line(s) of evidence to support the read-across outcome. While the available weight of evidence for BB17 suggests its safety in terms of genotoxicity, this is not the case for BB16. More details are provided in the SCCS overall comment on the genotoxicity of BB16.

SCCS overall comment on genotoxicity of Basic Brown 16

The analysis of all batches of Basic Brown 16 tested in different toxicological studies including genotoxicity testing indicates fluctuations in the impurity profiles of the test substance. In evaluating the genotoxicity potential of BB16, the SCCS took into consideration collectively all the available studies so far. The summary of the analysis of the results is presented below:

- ***In vitro* gene mutation:** 2 positive studies on gene mutations in bacteria (Ames test) and 2 positive studies on gene mutations in mammalian cells (1 study at the *tk* locus and 1 study on L5178Y cells at the *hprt* locus with the new batch VW230921A) and one negative study of limited reliability on V79 mouse fibroblasts at the *tk* locus.
- ***In vitro* chromosomal aberrations:** 1 positive study on chromosomal aberrations on V79 cells, 1 positive study on micronuclei induction on V79 cells and 1 equivocal micronucleus test on human lymphocytes with the new batch VW230921A.
- ***In vivo* micronucleus** tests on mice: 1 negative and 1 with inadequate result.

Considering all these data together, the SCCS is of the opinion that:

- the potential of Basic Brown 16 to induce gene mutations in bacteria and mammalian cells has been confirmed with the new study on L5178Y cells at the *hprt* locus with batch VW230921A, and the negative result of 1 Comet assay on 3D skin model analysed in the previous SCCS Opinion cannot overrule these positive *in vitro* results,
- the new study on chromosomal aberrations in human lymphocytes with batch VW230921A shows equivocal results and does not provide convincing evidence to exclude *in vitro* chromosomal aberration potential of Basic Brown 16,
- the *in vivo* results on micronucleus tests may collectively suggest evidence for the lack of potential of Basic Brown 16 to induce chromosomal aberrations *in vivo*; however, the studies were performed with older batches of the substance that may not be representative for the specification proposed for the global market material.

In silico analysis conducted by the SCCS (not included in this Opinion) indicates that:

- Basic Brown 16 contains structural alerts associated with mutagenicity: aromatic amine and aromatic diazo groups,
- although the reliability of the predictions is low because of the lack of similar substances to Basic Brown 16, all four QSAR models contained within the VEGA QSAR platform indicated the mutagenicity potential of Basic Brown 16,
- although Basic Brown 17 shares a structural similarity to Basic Brown 16, the read-across from a single analogue can only be accepted if it is supported by other lines of evidence. Also, both substances differ in terms of the weight of evidence for genotoxicity potential, with the overall WoE suggesting safety in terms of genotoxicity of Basic Brown 17 but not of Basic Brown 16.

Hence, the SCCS considers that a concern for the potential of Basic Brown 16 (including the new batch VW230921A) to induce gene mutations and chromosomal aberrations remains.

3.3.7. Carcinogenicity

SCCS comment

As genotoxicity of Basic Brown 16 cannot be excluded, the SCCS considers that the concerns for carcinogenicity cannot be ruled out.

3.3.8. Reproductive toxicity

From SCCS/1496/12:

"Neither embryotoxic nor teratogenic effects have been found in rats under the study conditions.

The NOAEL both for maternal and embryo foetal toxicity was considered to be 90 mg/kg bw/day.

As the dye content of the test substance is only 65.7%, the NOAEL value is re-calculated to 59 mg/kg bw/day."

3.3.9. Toxicokinetics

3.3.9.1. Toxicokinetics in laboratory animals

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3.3.9.2. Toxicokinetics in humans

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3.3.10. Photo-induced toxicity**3.3.10.1. Phototoxicity / photo-irritation and photosensitisation**

/

3.3.10.2. Photomutagenicity / photoclastogenicity

/

3.3.11. Human data

/

3.3.12. Special investigations

/

3.3.13. Safety evaluation (including calculation of the MoS)**From the Applicant**

Margin of safety calculation is based on dermal absorption of the test substance, considering the absorbable substance, considering the exposure to the substance and the Point of Departure.

The results are obtained from substance tested at 2% dissolved in water. The mean + 2 SD ($5.15 + 2 \times 2.32 = 9.79 \mu\text{g}/\text{cm}^2$) is used for calculating the MOS.

The NOAEL of 33 mg/kg bw/day from the 90-d oral study is adjusted considering a 50% oral absorption factor.

CALCULATION OF THE MARGIN OF SAFETY

(..... conditions)

(.....% formulation, on head concentration%)

Absorption through the skin	A	= 9.79 $\mu\text{g}/\text{cm}^2$
Skin Area surface	SAS	= 580 cm^2
Dermal absorption per treatment	SAS x A x 0.001	= 5.68 mg
Typical body weight of human		= 60 kg
Systemic exposure dose (SED)	SAS x A x 0.001/60	= 0.095 mg/kg bw
No observed adverse effect level (90-day, oral, rat)	NOAEL	= 33 mg/kg bw/d
Default oral absorption		= 50%
Bioavailability 50%*		= 16.5 mg/kg bw/d

Margin of Safety**adjusted NOAEL/SED = 174.4**

* standard procedure according to the SCCS's Notes of Guidance for the testing of cosmetic ingredients and their safety evaluation.

The Margin of Safety for BASIC BROWN 16 is 174 for the expected use.

Conclusion by the Applicant

BASIC BROWN 16 is non acutely toxic. It is not irritant for the skin and it is a moderate skin sensitizer. A NOAEL of 33 mg/kg bw/d has been identified in a 90-d oral study, it covers the NOAEL of 59 mg/kg bw/d in the teratogenicity study. This allows the calculation of a safe margin of safety for the expected use as hair dye at 2%.

It is noteworthy that the SCCS evaluated a similar substance (BB17) in 2014 and concluded it to be safe for use in non-oxidative hair dye formulations. The same positive effects were limited to bacteria models. Some limitation in the tests in mammalian systems were observed. It was not considered mutagenic in an *in vitro* micronucleus test, and negative in a micronucleus test *in vitro*, predictive for both clastogenicity and mutagenicity.

It seems that very similar genotoxicity profiles are observed for both BB17 and BB16.

The potential for BB16 to be genotoxic in humans has been widely investigated, in bacterial and mammalian test systems, in a human tissue skin model system and an *in vivo* study in the mouse. Positivity appear bacterial system, with metabolic activation, while mammalian models show some inconsistency between cell lines, gene loci, and study endpoints. The *in vitro* mammalian cells show equivocal or positive results are technically deficient or show inconsistencies that are incoherent, or no consistent dose response effects, in the last hprt assay. The reliable *in vitro* mammalian cell and tissue model assay (Comet assay in reconstructed human skin), and the *in vivo* study (micronucleus), show negative results. These results are relevant to consider the safe use of BB16, used as rinse-off hair dye.

SCCS comment

Based on the currently available information, it is not possible for the SCCS to exclude the mutagenicity potential of Basic Brown 16. The calculation of MoS has not been performed because a conclusion on the overall safety of Basic Brown 16 cannot be drawn without addressing the concern over its genotoxicity.

3.3.14. Discussion***Physicochemical properties***

Basic Brown 16 is used as a direct dye for hair colouring products without mixing with an oxidising agent (e.g. hydrogen peroxide). The final concentration on the head of Basic Brown 16 can be up to 2.0%.

In the current submission, the comparative analysis of purity and chemical composition was provided for 5 batches of Basic Brown 16: 57861/2, KS6054, 95424101, 12/13 and VW230921A.

While the main constituent and impurities of Basic Brown 16 have been more precisely characterised in batch 95424101, the overall impurity content is higher in the proposed specification for global market material compared to the batch used in the genotoxicity studies in submission V ter. (VW230921A). The increased impurity levels (up to 2.8%-area) raise uncertainty in regard to applicability of the provided toxicological data. Therefore, without further justification, the toxicological studies on VW230921A cannot be considered representative of the material covered under the proposed specifications.

General toxicity

Taken from previous SCCS Opinion SCCS/1496/12:

The acute median lethal oral dose (LD₅₀) of Basic Brown 16 was reported to be between 2 and 4 g/kg bw.

In a 90-day rat study, the NOAEL was considered to be 50 mg/kg bw/day, based on decreased body weight gain and some neurotoxic symptoms in both sexes. This value was re-calculated to 33 mg/kg bw/day based on the dye content of the test substance.

The NOAEL for both maternal and embryo-foetal toxicity was considered to be 90 mg/kg bw/day. This value was re-calculated to 59 mg/kg bw/day based on the dye content of the test substance.

Irritation/sensitisation

Taken from previous SCCS Opinion SCCS/1496/12:

Basic Brown 16 was considered not to be irritating to rabbit skin. It showed irritant potential to the rabbit eye.

Basic Brown 16 was a moderate skin sensitizer under the conditions of the test. An EC3 value of 12.2% was derived.

Dermal absorption

Taken from previous SCCS Opinion SCCS/1496/12:

The amount considered to be absorbed as direct dye formulation was 5.15 µg/cm² (range 2.19 to 9.12) or 1.42% of the applied dose (range 0.58% to 2.91%). At 2% dissolved in water, the amount considered to be absorbed was 4.57 µg/cm² (range 3.76 to 5.80) or 1.08% of the applied dose (range 0.89% to 1.38). As too few test chambers (8 chambers from 2 donors) the mean + 2 SD (5.15 + 2 × 2.32 = 9.79 µg/cm²) may be used for calculating the MOS.

Mutagenicity

The SCCS assessed all available studies in the current and previous submissions on mutagenicity of Basic Brown 16 and is of the opinion that a concern for the potential of Basic Brown 16 (including the new batch VW230921A) to induce gene mutations and chromosomal aberrations remains.

Carcinogenicity

As genotoxicity of Basic Brown 16 cannot be excluded, the SCCS considers that the concerns for carcinogenicity cannot be ruled out.

Toxicokinetics

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Human data

/

4. CONCLUSION

1. *In light of the data provided, does the SCCS consider Basic Brown 16 safe, when used as an ingredient at 2 % in non-oxidative hair dye formulations?*

Having assessed the data provided, the SCCS does not consider the use of Basic Brown 16 safe when used as an ingredient in non-oxidative hair dye formulations because the weight of evidence indicates its potential for mutagenicity.

2. *Does the SCCS have any further scientific concerns with regard to the use of Basic Brown 16 in cosmetic products?*

/

5. MINORITY OPINION

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6. REFERENCES

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7. GLOSSARY OF TERMS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – Appendix 15 - from page 158

8. LIST OF ABBREVIATIONS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – Appendix 15 - from page 158

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ANNEX 1. The assessment of all available studies on genotoxicity/mutagenicity of Basic Brown 16 by the SCCS with indication of differences in the assessments by the Applicant and the SCCS.

Applicant's evaluation summary					The SCCS evaluation summary					
Study Type	Result <i>In vitro</i> -S9	Result <i>In vitro</i> +S9	Result <i>In vivo</i>	Comment	Date of the study/ Submission#, SCCS Opinion	Batch/ purity	OECD TG#	Result	Comment	Reference
Ames Test	Negative	Positive	NA	Reproducible but generally weak responses in TA98 and TA1537 with pre-incubation.	Mar-2005 Subm III, 2008	57861/2 99.5% HPLC	471 GLP	Positive	Induction of mutations in TA98 and TA1537 (+S9)	Sokolowski A. (2005)
Ames Test					Sept-1985 Subm II, 2003	Arianor Mahogany KS 6024 Purity not provided	471	Positive	Induction of mutations in TA98 (+S9), in TA1537 (-/+S9), in TA1538 (+S9)	Wallat S. (1985)
Chromosome aberrations test					Mar-2001 Subm II, 2003	Arianor Mahogany Lot 7 Henkel Code No: SAT 000917 >94% HPLC	473	Positive	Clearly induced clastogenic effects -S9	Czich A. (2001)
Micronucleus V79 cells	Equivocal	Positive	NA	Study may be unreliable because cytotoxicity was not evaluated correctly.	Aug-2005 Subm III, 2008	57861/2 Henkel Code: SAT 050020 99.5% HPLC	487 GLP	Positive	Genotoxic after 4h treatment + 20h recovery	Kunz S. (2005)
Micronucleus Lymphocytes	Negative	Negative	NA	A modern study with human relevant cells that have some metabolic competence provided clear negative results.	Dec-2022 Subm V ter.	VW230921 A 99.6% HPLC	487 Non-GLP	Equivocal	After 3 h+S9 and 24 h-S9 exposure slight, but significantly increased MN frequency was observed at the lowest and	Cariou O. (2022)

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										highest concentrations tested	
	V79 hprt	Negative	Negative	NA	Study considered to be limited by SCCS because cytotoxicity was not evaluated correctly. However, the maximum dose +S9 exceeded the prescribed maximum, and the maximum dose -S9 was comparable with the hprt assay in L5178Y cells. The study is therefore reliable.	Jul-1991 Subm II, 2003	Arianor Mahogany KS 6024, Williams 73.1% (as chloride)	476 GLP	Limited value	Value is limited as toxicity assessment is restricted to cloning efficiency after the selection period only.	Michalke H. (1991)
	L5178Y hprt	Positive	Positive	NA	The positive result in the absence of S9 is unexpected because BB16 requires P450 activation and L5178Y cells are deficient in P450. The dose response relationship was not clear in both exposure conditions and both experiments.	Jul-2024 Subm V ter. 2024	VW230921 A 90.0% ¹ H-NMR	476 GLP	Positive	Positive –/+S9	Nisar R. (2024)
	L5178Y tk	Equivocal	Negative	NA	Results with S9 contradict those for the same cell line at the hprt locus. The result without S9 is unexpected considering the metabolic deficiency of the cells. In the	May-2005 Subm III, 2008	57861/2 Henkel Code: SAT 050020 99.5% HPLC	476 GLP	Positive	Positive –S9	Wollny H.E. (2005)

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					absence of S9 the cytotoxicity was at the upper limit of acceptability.						
	Epiderm Comet assay	Negative	NA	NA	A recent validation paper for the methodology supports the validity of this tissue relevant assay.	Jul-2011 Subm IV, 2012	batch 1: 8-778 (concentration range-finder experiment) batch 2: 95424101 Purity: batch 1: >95% (concentration range-finder experiment) batch 2: >99.0% (HPLC at 254 nm)	/	Negative	Negative	Pfuhler S. (2012)
	Micronucleus in mice					Oct-1987 Subm II, 2003	Arianor Mahogany ** Not given	474	Inadequate	Only one dose was tested (3000 mg/kg bw by gavage), ratio of normochromatic to polychromatic erythrocytes was not different from the control what may indicate lack of distribution to bone marrow.	Banduhn N. (1987)
	Micronucleus in mice	NA	NA	Negative	This is a robust study and shows no evidence of clastogenic or aneugenic potential <i>in vivo</i> .	Sept-2005 Subm III, 2008	57861/2 ** Henkel Code: SAT 050020 Purity: 99.5%	474	Negative	Intraperitoneal, up to 75 mg/kg Acute signs observed in pre-experiment and dark red-brown urine indicate systemic availability.	Honarvar N. (2005)

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Other endpoints										
DNA Damage and Repair- Unscheduled DNA Synthesis- Mammalian Cells <i>In vitro</i>					1983 Subm II, 2003	Not given	482	Inadequate	HeLa cells used	Wingen F <i>et al.</i> (1983)

* Applicant's comment on purity: The given value for the purity of the test substance used in the Ames test (Batch 57861/2) should become 99.5 area % (HPLC) as correctly given in the batch-related analytical report R 0500242 (Ref 2 of SMII in the corresponding SCCS opinion). The mentioned 98.8% area incorrectly referenced in the study report of this gene mutation study. The value of 98.8 area % is reported in other studies as well (e.g. MLA-tk 2005).

** The SCCS comment on the batches used: the studies were performed with older batches of the substance that may not be representative for the specification proposed for the global market material in the current submission.
